

D S Hewavithana

ID – 37544

Module Code – CN201.3

Assignment Title - Router Configuration and
Routing Protocols using Cisco Packet Tracer

Task 1

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname kandyNSBM_37544
kandyNSBM_37544(config)#line console 0
kandyNSBM_37544(config-line)#password NSBM
kandyNSBM_37544(config-line)#login
kandyNSBM_37544(config-line)#exit
kandyNSBM_37544(config)#service password-encryption
kandyNSBM_37544(config)#end
kandyNSBM_37544#
%SYS-5-CONFIG_I: Configured from console by console

kandyNSBM_37544#write memory
Building configuration...
[OK]
```

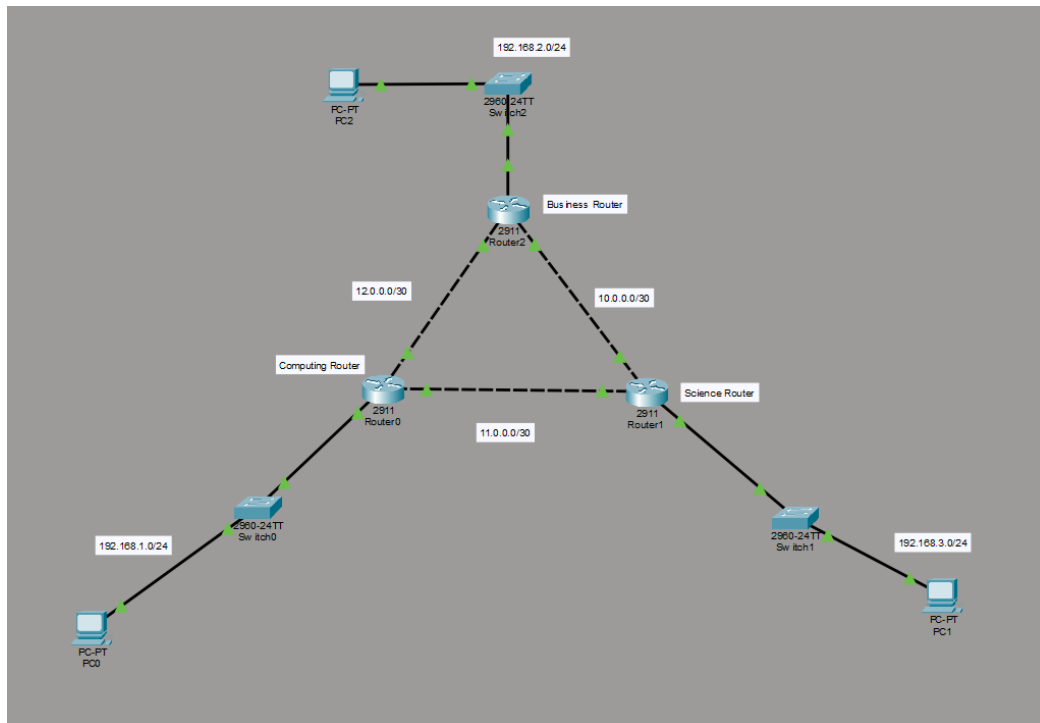
Press RETURN to get started.

User Access Verification

Password:

kandyNSBM_37544>

Task 2



Router 0 (Computing Router)

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname ComputingRouter
ComputingRouter(config)#interface GigabitEthernet0/0
ComputingRouter(config-if)#ip address 192.168.1.1 255.255.255.0
ComputingRouter(config-if)#no shutdown

ComputingRouter(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

ComputingRouter(config-if)#interface GigabitEthernet0/1
ComputingRouter(config-if)#ip address 12.0.0.1 255.255.255.252
ComputingRouter(config-if)#no shutdown

ComputingRouter(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

ComputingRouter(config-if)#interface GigabitEthernet0/2
ComputingRouter(config-if)#ip address 11.0.0.2 255.255.255.252
ComputingRouter(config-if)#no shutdown

ComputingRouter(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
exit
```

Router 1 (Science Router)

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname ScienceRouter
ScienceRouter(config)#interface GigabitEthernet0/0
ScienceRouter(config-if)#ip address 192.168.3.1 255.255.255.0
ScienceRouter(config-if)#no shutdown
ScienceRouter(config-if)#interface GigabitEthernet0/1
ScienceRouter(config-if)#ip address 10.0.0.2 255.255.255.252
ScienceRouter(config-if)#no shutdown

ScienceRouter(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

ScienceRouter(config-if)#interface GigabitEthernet0/2
ScienceRouter(config-if)#ip address 11.0.0.1 255.255.255.252
ScienceRouter(config-if)#no shutdown

ScienceRouter(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
exit
```

Router 2(Business Router)

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname BusinessRouter
BusinessRouter(config)#interface GigabitEthernet0/0
BusinessRouter(config-if)#ip address 192.168.2.1 255.255.255.0
BusinessRouter(config-if)#no shutdown

BusinessRouter(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

BusinessRouter(config-if)#interface GigabitEthernet0/1
BusinessRouter(config-if)#ip address 12.0.0.2 255.255.255.252
BusinessRouter(config-if)#no shutdown

BusinessRouter(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

BusinessRouter(config-if)#interface GigabitEthernet0/2
BusinessRouter(config-if)#ip address 10.0.0.1 255.255.255.252
BusinessRouter(config-if)#no shutdown

BusinessRouter(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
exit
BusinessRouter(config)#exit
BusinessRouter#
%SYS-5-CONFIG_I: Configured from console by console
exit
```

Routing

```
ComputingRouter(config)#ip route 192.168.2.0 255.255.255.0 12.0.0.2
ComputingRouter(config)#ip route 192.168.3.0 255.255.255.0 11.0.0.1
ComputingRouter(config)#end
ComputingRouter#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
ComputingRouter#
```

```
BusinessRouter>enable
BusinessRouter#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
BusinessRouter(config)#ip route 192.168.1.0 255.255.255.0 12.0.0.1
BusinessRouter(config)#ip route 192.168.3.0 255.255.255.0 10.0.0.2
BusinessRouter(config)#end
BusinessRouter#
%SYS-5-CONFIG_I: Configured from console by console

BusinessRouter#write memory
Building configuration...
[OK]
```

```
ScienceRouter>enable
ScienceRouter#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ScienceRouter(config)#ip route 192.168.1.0 255.255.255.0 11.0.0.2
ScienceRouter(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.1
ScienceRouter(config)#end
ScienceRouter#
%SYS-5-CONFIG_I: Configured from console by console

ScienceRouter#write memory
Building configuration...
[OK]
ScienceRouter#
```

Routing Tables

Computing Router

```
ComputingRouter#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       11.0.0.0/30 is directly connected, GigabitEthernet0/2
L       11.0.0.2/32 is directly connected, GigabitEthernet0/2
    12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       12.0.0.0/30 is directly connected, GigabitEthernet0/1
L       12.0.0.1/32 is directly connected, GigabitEthernet0/1
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
S       192.168.2.0/24 [1/0] via 12.0.0.2
S       192.168.3.0/24 [1/0] via 11.0.0.1
```

Business Router

```
BusinessRouter>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/2
L       10.0.0.1/32 is directly connected, GigabitEthernet0/2
    12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       12.0.0.0/30 is directly connected, GigabitEthernet0/1
L       12.0.0.2/32 is directly connected, GigabitEthernet0/1
S       192.168.1.0/24 [1/0] via 12.0.0.1
    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0
L       192.168.2.1/32 is directly connected, GigabitEthernet0/0
S       192.168.3.0/24 [1/0] via 10.0.0.2
```

Science Router

```
ScienceRouter>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.2/32 is directly connected, GigabitEthernet0/1
    11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       11.0.0.0/30 is directly connected, GigabitEthernet0/2
L       11.0.0.1/32 is directly connected, GigabitEthernet0/2
S       192.168.1.0/24 [1/0] via 11.0.0.2
S       192.168.2.0/24 [1/0] via 10.0.0.1
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/0
L       192.168.3.1/32 is directly connected, GigabitEthernet0/0
```

Ping Tests

Computing to Business Ping

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Computing to Science Ping

```
Pinging 192.168.3.10 with 32 bytes of data:

Reply from 192.168.3.10: bytes=32 time<1ms TTL=126
Reply from 192.168.3.10: bytes=32 time<1ms TTL=126
Reply from 192.168.3.10: bytes=32 time<1ms TTL=126
Reply from 192.168.3.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.3.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Business to Science Ping

```
C:\>ping 192.168.3.10

Pinging 192.168.3.10 with 32 bytes of data:

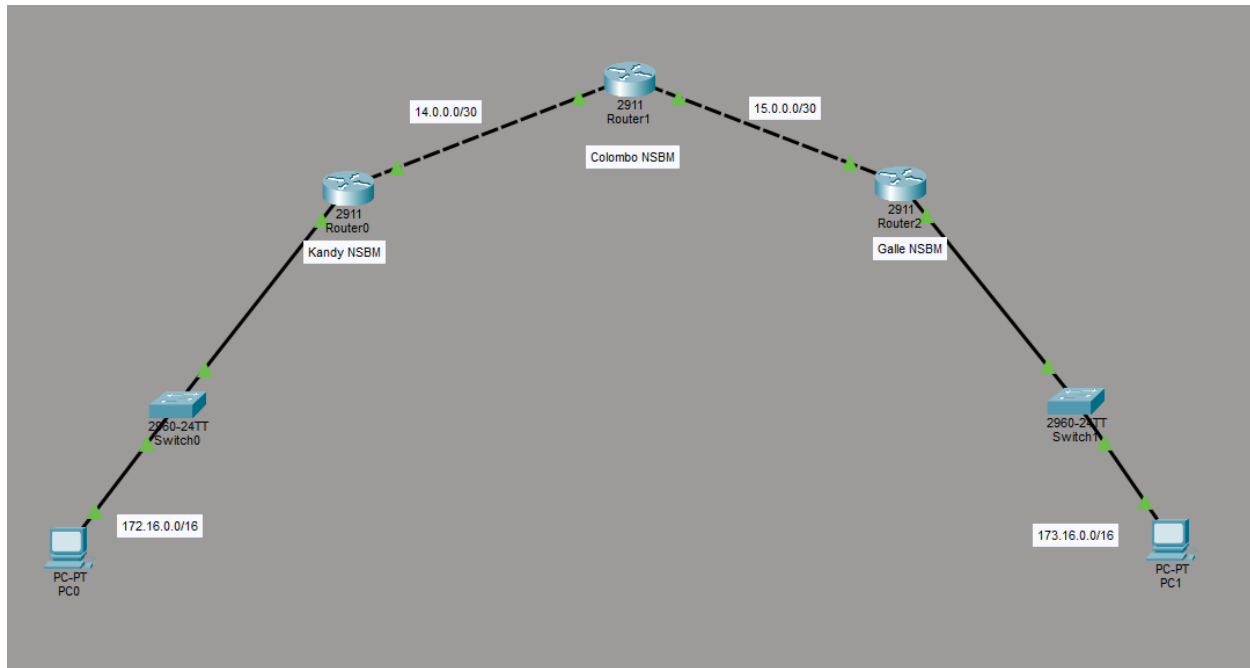
Reply from 192.168.3.10: bytes=32 time<1ms TTL=126
Reply from 192.168.3.10: bytes=32 time<1ms TTL=126
Reply from 192.168.3.10: bytes=32 time<1ms TTL=126
Reply from 192.168.3.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.3.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Some Commands Used

- “interface GigabitEthernet0/0” – used to select a specific port on a router
- “ip address 192.168.2.1 255.255.255.0” – assigns an IP address with the specified subnet mask to the selected port
- “no shutdown” – turns on the specified port
- “ip route 192.168.1.0 255.255.255.0 11.0.0.2” – specifies that if data needs to be transmitted into the 192.168.1.0 with the subnet 255.255.255.0 it should be sent via the 11.0.0.2 network

Task 3



```
KandyNSBM#enable
KandyNSBM#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
KandyNSBM(config)#ip address 172.16.0.1 255.255.0.0
^
% Invalid input detected at '^' marker.

KandyNSBM(config)#interface GigabitEthernet0/0
KandyNSBM(config-if)#ip address 172.16.0.1 255.255.0.0
KandyNSBM(config-if)#no shutdown

KandyNSBM(config-if)#
%LINK-S-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

KandyNSBM>enable
KandyNSBM#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
KandyNSBM(config)#interface GigabitEthernet0/1
KandyNSBM(config-if)#ip address 14.0.0.1 255.255.255.252
KandyNSBM(config-if)#no shutdown

KandyNSBM(config-if)#
%LINK-S-CHANGED: Interface GigabitEthernet0/1, changed state to up
exit
```

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname ColomboNSBM
ColomboNSBM(config)#interface GigabitEthernet0/1
ColomboNSBM(config-if)#ip address 14.0.0.2 255.255.255.252
ColomboNSBM(config-if)#no shutdown
```

```
ColomboNSBM(config)#interface GigabitEthernet0/2
ColomboNSBM(config-if)#ip address 15.0.0.1 255.255.255.252
ColomboNSBM(config-if)#no shutdown
```

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname GalleNSBM
GalleNSBM(config)#interface GigabitEthernet0/0
GalleNSBM(config-if)#ip address 173.16.0.0. 255.255.0.0
      ^
% Invalid input detected at '^' marker.

GalleNSBM(config-if)#
GalleNSBM(config-if)#ip address 173.16.0.1 255.255.0.0
GalleNSBM(config-if)#no shutdown
```

```
GalleNSBM(config)#interface GigabitEthernet0/1
GalleNSBM(config-if)#ip address 15.0.0.2 255.255.255.252
GalleNSBM(config-if)#no shutdown
```

Configuration Commands

```
KandyNSBM>enable
KandyNSBM#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
KandyNSBM(config)#router rip
KandyNSBM(config-router)#version 2
KandyNSBM(config-router)#no auto-summary
KandyNSBM(config-router)#network 172.16.0.0
KandyNSBM(config-router)#network 14.0.0.0
```

```
ColomboNSBM(config)#router rip
ColomboNSBM(config-router)#version 2
ColomboNSBM(config-router)#no auto-summary
ColomboNSBM(config-router)#network 14.0.0.0
ColomboNSBM(config-router)#network 15.0.0.0
^
% Invalid input detected at '^' marker.
ColomboNSBM(config-router)#network 15.0.0.0
```

```
GalleNSBM(config)#router rip
GalleNSBM(config-router)#version 2
GalleNSBM(config-router)#no auto-summary
^
% Invalid input detected at '^' marker.
GalleNSBM(config-router)#no auto-summary
GalleNSBM(config-router)#network 173.16.0.0
GalleNSBM(config-router)#network 15.0.0.0
```

RIP Ping Outputs

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 173.16.0.10

Pinging 173.16.0.10 with 32 bytes of data:

Request timed out.
Reply from 173.16.0.10: bytes=32 time<1ms TTL=125
Reply from 173.16.0.10: bytes=32 time<1ms TTL=125
Reply from 173.16.0.10: bytes=32 time<1ms TTL=125

Ping statistics for 173.16.0.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
ColomboNSBM#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    14.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       14.0.0.0/30 is directly connected, GigabitEthernet0/1
L       14.0.0.2/32 is directly connected, GigabitEthernet0/1
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       15.0.0.0/30 is directly connected, GigabitEthernet0/2
L       15.0.0.1/32 is directly connected, GigabitEthernet0/2
R       172.16.0.0/16 [120/1] via 14.0.0.1, 00:00:23, GigabitEthernet0/1
R       173.16.0.0/16 [120/1] via 15.0.0.2, 00:00:04, GigabitEthernet0/2
```

```
KandyNSBM>enable
KandyNSBM#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    14.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       14.0.0.0/30 is directly connected, GigabitEthernet0/1
L       14.0.0.1/32 is directly connected, GigabitEthernet0/1
    15.0.0.0/30 is subnetted, 1 subnets
R       15.0.0.0/30 [120/1] via 14.0.0.2, 00:00:06, GigabitEthernet0/1
    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/16 is directly connected, GigabitEthernet0/0
L       172.16.0.1/32 is directly connected, GigabitEthernet0/0
R       173.16.0.0/16 [120/2] via 14.0.0.2, 00:00:06, GigabitEthernet0/1
```

```

show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    14.0.0.0/30 is subnetted, 1 subnets
R       14.0.0.0/30 [120/1] via 15.0.0.1, 00:00:07, GigabitEthernet0/1
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       15.0.0.0/30 is directly connected, GigabitEthernet0/1
L       15.0.0.2/32 is directly connected, GigabitEthernet0/1
R       172.16.0.0/16 [120/2] via 15.0.0.1, 00:00:07, GigabitEthernet0/1
    173.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       173.16.0.0/16 is directly connected, GigabitEthernet0/0
L       173.16.0.1/32 is directly connected, GigabitEthernet0/0

```

Removing RIP

```

KandyNSBM#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
KandyNSBM(config)#no router rip

ColomboNSBM#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ColomboNSBM(config)#no router rip

GalleNSBM#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
GalleNSBM(config)#no router rip

```

Setting up EIGRP Routing

```
KandyNSBM(config)#router eigrp 10
KandyNSBM(config-router)#no auto-summary
KandyNSBM(config-router)#network 172.16.0.0
KandyNSBM(config-router)#network 14.0.0.0

ColomboNSBM(config)#router eigrp 10
ColomboNSBM(config-router)#no auto-summary
ColomboNSBM(config-router)#network 14.0.0.0
ColomboNSBM(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 10: Neighbor 14.0.0.1 (GigabitEthernet0/1) is
up: new adjacency
ColomboNSBM(config-router)#network 15.0.0.0

GalleNSBM(config)#router eigrp 10
GalleNSBM(config-router)#no auto-summary
GalleNSBM(config-router)#network 173.16.0.0
GalleNSBM(config-router)#network 15.0.0.0
GalleNSBM(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 10: Neighbor 15.0.0.1 (GigabitEthernet0/1) is
up: new adjacency
```

Ping Outputs

```
C:\>ping 172.16.0.10

Pinging 172.16.0.10 with 32 bytes of data:

Reply from 172.16.0.10: bytes=32 time=3ms TTL=128
Reply from 172.16.0.10: bytes=32 time=1ms TTL=128
Reply from 172.16.0.10: bytes=32 time=3ms TTL=128
Reply from 172.16.0.10: bytes=32 time=3ms TTL=128

Ping statistics for 172.16.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms
```

```
C:\>ping 172.16.0.10

Pinging 172.16.0.10 with 32 bytes of data:

Reply from 172.16.0.10: bytes=32 time<1ms TTL=125
Reply from 172.16.0.10: bytes=32 time<1ms TTL=125
Reply from 172.16.0.10: bytes=32 time<1ms TTL=125
Reply from 172.16.0.10: bytes=32 time<1ms TTL=125

Ping statistics for 172.16.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Routing Table Outputs

```
KandyNSBM#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    14.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       14.0.0.0/30 is directly connected, GigabitEthernet0/1
L       14.0.0.1/32 is directly connected, GigabitEthernet0/1
    15.0.0.0/30 is subnetted, 1 subnets
D       15.0.0.0/30 [90/3072] via 14.0.0.2, 00:03:50, GigabitEthernet0/1
    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/16 is directly connected, GigabitEthernet0/0
L       172.16.0.1/32 is directly connected, GigabitEthernet0/0
D       173.16.0.0/16 [90/28672] via 14.0.0.2, 00:02:47, GigabitEthernet0/1
```

```
ColomboNSBM#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    14.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       14.0.0.0/30 is directly connected, GigabitEthernet0/1
L       14.0.0.2/32 is directly connected, GigabitEthernet0/1
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       15.0.0.0/30 is directly connected, GigabitEthernet0/2
L       15.0.0.1/32 is directly connected, GigabitEthernet0/2
D       172.16.0.0/16 [90/28416] via 14.0.0.1, 00:05:49, GigabitEthernet0/1
D       173.16.0.0/16 [90/28416] via 15.0.0.2, 00:03:31, GigabitEthernet0/2
```

```
GalleNSBM#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

  14.0.0.0/30 is subnetted, 1 subnets
D    14.0.0.0/30 [90/3072] via 15.0.0.1, 00:03:56, GigabitEthernet0/1
  15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    15.0.0.0/30 is directly connected, GigabitEthernet0/1
L    15.0.0.2/32 is directly connected, GigabitEthernet0/1
D    172.16.0.0/16 [90/28672] via 15.0.0.1, 00:03:56, GigabitEthernet0/1
  173.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    173.16.0.0/16 is directly connected, GigabitEthernet0/0
L    173.16.0.1/32 is directly connected, GigabitEthernet0/0
```

Some of the commands used

- “router rip” – instructs the router to stop using static routing and
- “router eigrp 10” – the command initializes the EIGRP routing process and gives 10 as the Autonomous ID
- “version 2” – directs to select version 2 in RIP (version 2 uses subnet masks)
- “network 14.0.0.0” – this command executes two actions. First it tells the routing protocol to check its physical interfaces and find the port that matches that network address and advertise it to its neighbors. Second, it tells the router to start sending and listening for routing updates out of that specific physical interface port
- “no router rip” – it deletes the RIP configuration

